

CS 559 Deep Learning Term Project: Adaptive Speech Recognition via Deep Meta-Learning

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Motivation:

- Real-world audio is noisy and diverse
- No single speech-to-text (STT) model is universally optimal
- Running multiple models in real time is infeasible

Goal: Build an adaptive system that selects the most suitable STT model

Solution:

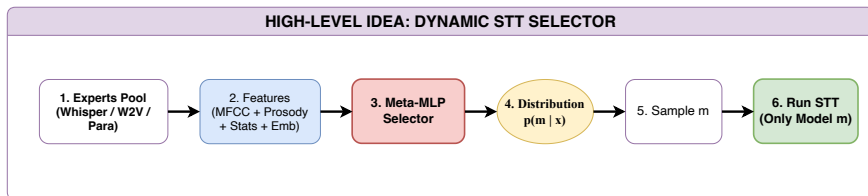


Figure: High-level concept of the adaptive selector.

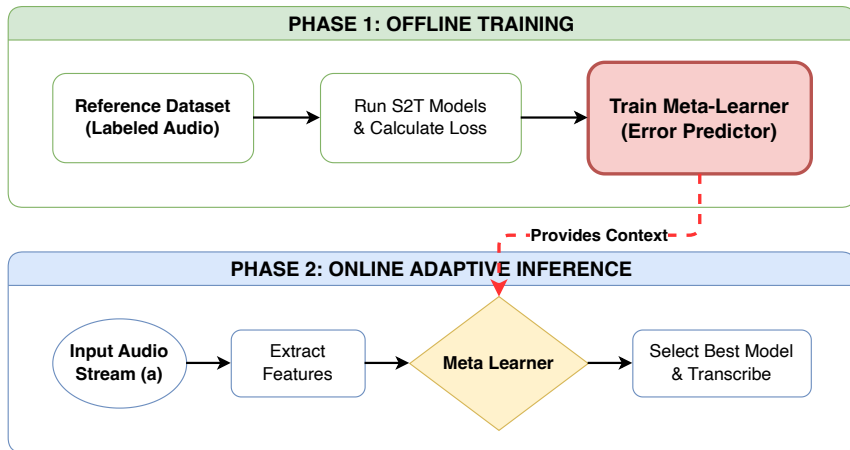


Figure: Two-phased system architecture.

Offline Training Phase

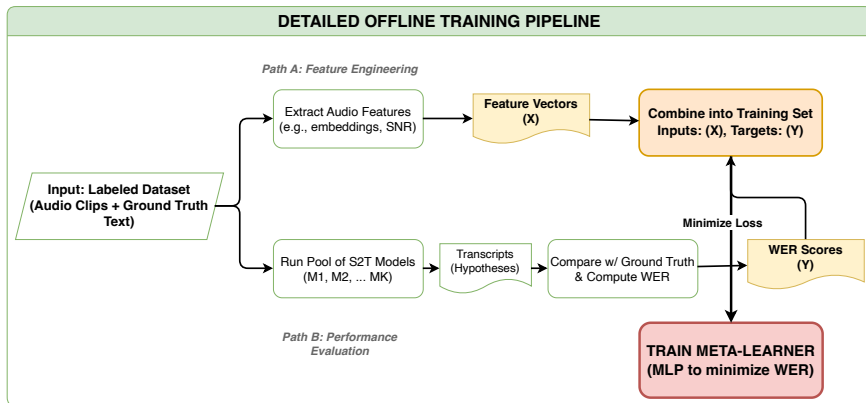


Figure: Offline training phase: The meta-learner learns to assign high probability ($w_{n,m}$) to the expert with the lowest Word Error Rate (WER).

$$\text{WER} = \frac{S+D+I}{N}$$

$$L_n = \sum_m w_{n,m} \cdot \text{WER}_{n,m}$$

Input Features

- Spectral-Phonetic
- Prosodic
- Signal Complexity
- Neural Embeddings



Encoded Information

- Temporal structure
- Speaker identity
- Noise characteristics
- Semantic cues

The meta-learner leverages these diverse signals to predict the optimal expert.

Online Inference Phase

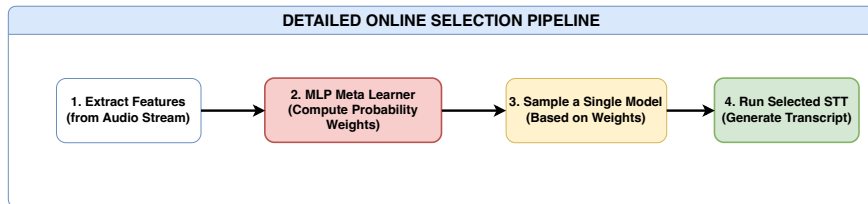


Figure: Online inference workflow: Features extracted from input audio are passed to the MLP meta-learner, which selects a single expert model to execute.

STT Expert Pool

- Whisper Large-V3 [1]
- NVIDIA Parakeet-TDT 0.6B [2]
- NVIDIA Canary-Qwen-2.5B [3]

Baselines & Ensembles

- Uniform Random Ensemble
- Weighted Random Ensemble
- **Oracle** (Theoretical Lower Bound)

Datasets

Evaluated on 3 diverse ASR domains:

1. LibriSpeech ASR [4]
2. Mozilla Common Voice 22.0 [5]
3. VoxPopuli [6]

[1] A. Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision," *Proc. ICML*, 2023.

[2] M. Sekoyan et al., "Efficient and High-Performance Models for Multilingual ASR and AST," *arXiv preprint*, 2025.

[3] NVIDIA, "Canary-Qwen-2.5B," Hugging Face repository, 2025.

[4] V. Panayotov et al., "Librispeech: an ASR corpus based on public domain audio books," *Proc. IEEE ICASSP*, 2015.

[5] R. Ardila et al., "Common Voice: A Massively-Multilingual Speech Corpus," *Proc. LREC*, 2020.

[6] C. Wang et al., "VoxPopuli: A Large-Scale Multilingual Speech Corpus," *Proc. ACL*, 2021.

We evaluated our method across three distinct domains to test generalization:

LibriSpeech [4]

Clean vs. Noisy

- ~1000h read speech.
- **test-clean:** High-quality, clear audio.
- **test-other:** Noisy, harder conditions.
- Standard ASR benchmark.

Common Voice [5]

Robustness Test

- Crowdsourced, diverse accents.
- Real-world noise & conditions.
- High speaker diversity.
- Tests generalization to "in-the-wild" data.

VoxPopuli [6]

Domain Shift

- Parliamentary speech (EU).
- Long-form, formal & spontaneous.
- Real-world acoustics (reverb).
- Evaluates domain-shifted performance.

[4] V. Panayotov et al., "Librispeech: An ASR Corpus Based on Public Domain Audio Books," *Proc. IEEE ICASSP*, 2015.

[5] R. Ardila et al., "Common Voice: A Massively-Multilingual Speech Corpus," *Proc. LREC*, 2020.

[6] C. Wang et al., "VoxPopuli: A Large-Scale Multilingual Speech Corpus," *Proc. ACL*, 2021.

Model Selection & Architecture Search



Experimented with three progressive architectures:

Architecture 1

3 Hidden Layers

$256 \rightarrow 128 \rightarrow 64$

Activation: **RELU**

Limited capacity

Architecture 2

4 Hidden Layers

$512 \rightarrow 256 \rightarrow 128 \rightarrow 64$

Activation: **GELU**

Smoother gradients

Architecture 3

5 Hidden Layers

$1024 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 64$

Activation: **GELU**

★ *Optimal balance* ★

Shared Configuration & Output

- **Regularization:** BatchNorm + Dropout applied to all architectures.
- **Training:** AdamW optimizer, mixed precision, gradient clipping.
- **Output:** Logits $z_n \rightarrow \text{Softmax} \rightarrow$ Probability vector w_n over experts.

Comparative Results



Model	LibriSpeech-Clean ^[4]	LibriSpeech-Other ^[4]	Common Voice ^[5]	VoxPopuli ^[6]
Oracle (Upper Bound)	0.0075	0.0157	0.0412	0.0352
Whisper Large-V3 ^[1]	0.0172	0.0382	0.0797	0.0968
Parakeet-TDT 0.6B ^[2]	0.0159	0.0351	0.0637	0.0605
Canary-Qwen 2.5B ^[3]	0.0226	0.0395	0.1003	0.0581
Uniform Random Ensemble	0.0187	0.0378	0.0835	0.0720
Weighted Random Ensemble	0.0171	0.0372	0.0720	0.0625
ADASTT (Ours)	0.0131	0.0283	0.0638	0.0493
<i>Relative Improv.</i>	<i>+13.5%</i>	<i>+19.4%</i>	<i>+11.4%</i>	<i>+15.2%</i>

Table: Word Error Rate (WER) across all datasets.

[1] A. Radford et al., "Robust Speech Recognition via Large-Scale Weak Supervision," *Proc. ICML*, 2023.

[2] M. Sekoyan et al., "Efficient and High-Performance Models for Multilingual ASR and AST," *arXiv preprint*, 2025.

[3] NVIDIA, "Canary-Qwen-2.5B," Hugging Face repository, 2025.

[4] V. Panayotov et al., "Librispeech: an ASR corpus based on public domain audio books," *Proc. IEEE ICASSP*, 2015.

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[6] C. Wang et al., "VoxPopuli: A Large-Scale Multilingual Speech Corpus," *Proc. ACL*, 2021.

Model	LibriSpeech-Clean ^[4]	LibriSpeech-Other ^[4]	Common Voice ^[5]	VoxPopuli ^[6]
Feature Ext.	0.1845	0.1249	0.1441	0.2737
Whisper Large-V3 ^[1]	0.6331	0.5803	0.3682	0.8019
Parakeet-TDT 0.6B ^[2]	0.0502	0.0491	0.0465	0.0530
Canary-Qwen 2.5B ^[3]	0.1506	0.1473	0.1395	0.1590
Uniform Random Ensemble	0.4625	0.3838	0.3288	0.6117
Weighted Random Ensemble	0.3523	0.2831	0.2668	0.4708
ADASTT (Ours)	0.6399	0.3445	0.2073	1.0184

Table: Average processing time (in seconds) per input segment.

- Feature extraction takes $\approx$ **0.12–0.28 s** per input across datasets.
- Meta-learner adds minimal overhead while improving WER

[1] A. Radford et al., “Robust Speech Recognition via Large-Scale Weak Supervision,” *Proc. ICML*, 2023.

[2] M. Sekoyan et al., “Efficient and High-Performance Models for Multilingual ASR and AST,” *arXiv preprint*, 2025.

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[6] C. Wang et al., “VoxPopuli: A Large-Scale Multilingual Speech Corpus,” *Proc. ACL*, 2021.

Impact of removing feature groups one by one (Error Rates)

Configuration	LS-Other ^[4]	Common Voice ^[5]	VoxPopuli ^[6]	Average
Full Model	0.0283	0.0638	0.0493	0.0471
w/o MFCC	0.0289	0.0651	0.0503	0.0481
w/o Neural Emb.	0.0323	0.1462	0.3087	0.1624
w/o Prosody	0.0294	0.0664	0.0513	0.0490
w/o Signal Comp.	0.0286	0.0657	0.0498	0.0480

Table: Ablation results showing the contribution of each feature group to overall performance.

- Removing neural embeddings causes the largest degradation.
- Other features provide smaller but consistent improvements.
- **Takeaway:** Importance is context-dependent!

[4] V. Panayotov et al., "Librispeech: an ASR corpus based on public domain audio books," *Proc. IEEE ICASSP*, 2015.

[5] R. Ardila et al., "Common Voice: A Massively-Multilingual Speech Corpus," *Proc. LREC*, 2020.

[6] C. Wang et al., "VoxPopuli: A Large-Scale Multilingual Speech Corpus," *Proc. ACL*, 2021.

- **Novel Method:** Adaptive, segment-level STT selection via deep meta-learning.
- **Performance:** Achieved **11.4–19.5%** relative WER reduction with negligible cost.
- **Robustness:** Validated across diverse domains (clean, noisy, accented speech).

Thank You!
Questions?